

An elementary quadratic bound for the last atom

Complete elementary proof; three independent mathematical reviews

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Let $w_q > 0$, $1 \leq q \leq K$, sum to one, and let $p_i(q) \in [0, 1]$, $1 \leq i \leq m$, be nondecreasing in q . Suppose

$$\sum_q w_q \prod_{i \in S} p_i(q) = \frac{1}{|S| + 1} \quad (S \subseteq \{1, \dots, m\}). \quad (1)$$

The empty-set equation is the normalization. No equal weights or rationality are assumed. Write $a_i = p_i(K)$.

Theorem 1. *For every $m \geq 1$,*

$$a_i \geq 1 - \frac{\pi^2}{(m+1)^2} \quad (1 \leq i \leq m), \quad w_K \leq \frac{200}{m^2}.$$

The second bound also holds for the total weight of all rows identical to the last vector. Consequently an equal-weight representation has $K \geq m^2/200$.

The proof uses only squarefree moment exactness, monotonicity, and ordinary Gaussian and Radau quadrature. In particular it gives a short independent proof of the exact quadratic per-die lower bound, without the endpoint localization estimates developed earlier in this project.

Lemma 1 (Paired thresholds). *Complete the coordinatewise ordered row vectors to a continuous nondecreasing path by adjoining the vectors 0 and 1 and linearly interpolating successive vectors. For distinct indices j, k and $t \in (0, 1)$, stop at the first time either coordinate reaches t . The resulting threshold values α, β satisfy*

$$\{\alpha, \beta\} = \{t, b\}, \quad 0 \leq b \leq t, \quad (p_j(q) - \alpha)(p_k(q) - \beta) \geq 0 \quad \text{for every original row } q.$$

Products of these factors on disjoint pairs remain nonnegative.

Proof. Both coordinates start at zero and end at one. At their first crossing of t , one equals t and the other is at most t . Every original row before the stopping point is coordinatewise below the threshold vector, and every row after it is coordinatewise above it. Thus the two differences have the same weak sign. The added path points have no probability weight and change none of the moment equations. $\square$

For every multi-affine polynomial F , equation (1) implies the exact diagonal identity

$$\mathbb{E}_w F(p_1, \dots, p_m) = \int_0^1 F(t, \dots, t) dt. \quad (2)$$

This is simply linearity applied to its squarefree monomials.

Lemma 2 (Quadrature facts). *The D -node Gaussian rule for Lebesgue measure on $[0, 1]$ has positive weights, is exact through degree $2D - 1$, and its largest node ξ_D satisfies*

$$1 - \xi_D \leq \frac{\pi^2}{4(D+1)^2}. \quad (3)$$

The $(e + 1)$ -node right Radau rule has nodes $t_1 < \dots < t_e < 1$ and 1, positive weights, and exactness through degree $2e$. Its endpoint weight and interior-node sum are

$$\omega_1 = \frac{1}{(e + 1)^2}, \quad \sum_{\ell=1}^e \frac{1}{1 - t_\ell} = \frac{e(e + 2)}{2}. \quad (4)$$

Proof. Let P_r be the usual Legendre polynomial with $P_r(1) = 1$. Orthogonality yields the Gaussian rule at the shifted roots of P_D : division by the node polynomial gives exactness, and applying the rule to squared cardinal polynomials gives positive weights. The three-term Legendre recurrence shows that its unshifted roots are the eigenvalues of the symmetric $D \times D$ Jacobi matrix with zero diagonal and off-diagonal entries $k/\sqrt{(2k - 1)(2k + 1)} \geq 1/2$. The positive sine eigenvector of the matrix with all off-diagonals $1/2$ gives, by the Rayleigh principle, largest eigenvalue at least $\cos(\pi/(D + 1))$. Shifting to $[0, 1]$ and using $1 - \cos x \leq x^2/2$ proves (3).

For Radau put

$$J_e(t) = \sum_{r=0}^e (2r + 1)P_r(2t - 1).$$

This is the reproducing kernel at 1 for degree- e polynomials: $\int_0^1 J_e(t)f(t)dt = f(1)$. Hence J_e is orthogonal to degree at most $e - 1$ for the positive weight $1 - t$, and has e distinct interior roots. Interpolation at these roots and 1 integrates every polynomial of degree at most $2e$: the division remainder has degree at most e , and the quotient term is $(t - 1)J_e(t)q(t)$ with $\deg q \leq e - 1$. Interior weights are positive by evaluating the rule on $(1 - t)(J_e(t)/(t - t_\ell))^2$. At the endpoint, exactness on J_e^2 and the reproducing identity give

$$\omega_1 J_e(1)^2 = \int_0^1 J_e(t)^2 dt = J_e(1) = (e + 1)^2.$$

Finally the elementary endpoint derivative formula for shifted Legendre polynomials is $(P_r(2t - 1))'|_{t=1} = r(r + 1)$. Therefore

$$\frac{J'_e(1)}{J_e(1)} = \frac{\sum_{r=0}^e (2r + 1)r(r + 1)}{(e + 1)^2} = \frac{e(e + 2)}{2}.$$

The left side is the logarithmic derivative over the roots, proving the second identity in (4). $\square$

Proof of Theorem 1. Fix i and put $d = \lfloor (m - 1)/2 \rfloor$, $D = d + 1$. Choose d disjoint pairs among the other $m - 1$ columns. For pair ℓ , apply Lemma 1 at the ℓ th node of the D -node Gaussian rule, omitting its largest node ξ_D . Let $T(p)$ be the product of the d paired factors and put

$$P(t) = T(t, \dots, t) = \prod_{\ell=1}^d (t - \xi_\ell)(t - b_\ell), \quad b_\ell \leq \xi_\ell < \xi_D.$$

Both $T(p(q))$ and $a_i - p_i(q)$ are nonnegative. They use disjoint columns, so their product is multi-affine, of degree $2d + 1$. By (2) and Gaussian exactness,

$$0 \leq \mathbb{E}_w[(a_i - p_i)T(p)] = \int_0^1 (a_i - t)P(t)dt = \omega_D(a_i - \xi_D)P(\xi_D).$$

Here $\omega_D > 0$ and $P(\xi_D) > 0$, including the empty-product case $d = 0$. Thus $a_i \geq \xi_D$ for every i . Since $D + 1 \geq (m + 1)/2$, equation (3) gives $1 - a_i \leq \delta := \pi^2/(m + 1)^2$.

For $m \geq 10$, put $e = \lfloor m/10 \rfloor$ and choose e disjoint pairs of columns. Apply the threshold lemma to these pairs at the e interior nodes of the right Radau rule. Again write T for the nonnegative multi-affine product and

$$P(t) = \prod_{\ell=1}^e (t - t_\ell)(t - b_\ell), \quad b_\ell \leq t_\ell.$$

The degree is $2e \leq m$. Radau exactness and (2) give

$$\mathbb{E}_w T(p) = \frac{P(1)}{(e+1)^2}. \quad (5)$$

Set $S = e(e+2)/2$. As $e \leq m/10$ and $e+2 \leq 3m/10$,

$$\delta S \leq \frac{3\pi^2}{200} < \frac{1}{4}.$$

In particular all threshold roots are strictly below every a_i . At the last row each paired factor, divided by its value at the all-one vector, is at least $(1 - \delta/(1 - t_\ell))^2$. The elementary inequality $\prod_j (1 - z_j) \geq 1 - \sum_j z_j$ for $0 \leq z_j \leq 1$ therefore gives

$$\frac{T(a)}{P(1)} \geq \prod_{\ell=1}^e \left(1 - \frac{\delta}{1 - t_\ell}\right)^2 \geq 1 - 2\delta S > \frac{1}{2}.$$

Consequently (5) implies $w_K \leq 2/(e+1)^2 \leq 200/m^2$. Identical terminal rows contribute the same value $T(a)$, giving the aggregate version. For $m < 10$ the asserted bound follows from $w_K \leq 1$. $\square$

Corollary 1 (Fair dice). *Every die in a permutation-fair family of $n \geq 2$ finite uniform dice has at least $(n-1)^2/200$ faces.*

Proof. Condition on the sorted faces of that die. The comparison probabilities of the other $m = n-1$ dice are monotone and satisfy (1). Each conditioning row has weight the reciprocal of its face count. Apply Theorem 1. $\square$

Only the mixed moment conditions (1) were used. Thus full permutation fairness can be weakened to uniform winning probability for this distinguished die on every subset containing it.